

# Paper Replication Report #041: Atmospheric Escape & Hydrodynamic Photo-Evaporation

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

## Abstract

We present a first-principles replication of Watson, Donahue, & Walker (1981) and Tian et al. (2005) modeling hydrodynamic atmospheric escape and EUV photo-evaporation mass loss rates  $\dot{M}_{\text{loss}}$  for terrestrial planets and hot Jupiters. Using our C++ solver engine, we evaluate energy-limited mass loss across EUV irradiation levels  $F_{\text{EUV}} \in [10^{-4}, 1.0] \text{ W/m}^2$ . Numerical mass loss rates match hydrodynamic upper-atmosphere expansion models with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

The energy-limited hydrodynamic atmospheric escape rate  $\dot{M}_{\text{loss}}$  driven by stellar extreme ultraviolet (EUV) flux  $F_{\text{EUV}}$  with heating efficiency  $\epsilon$  is given by:

$$\dot{M}_{\text{loss}} = \frac{\epsilon \pi F_{\text{EUV}} R_p^3}{GM_p} \quad (1)$$

## 2 Replication Results

The C++ solver target `//:hydrodynamic_escape_solver` computes atmospheric escape. Under early Sun EUV flux  $F_{\text{EUV}} = 0.1 \text{ W/m}^2$  (100× modern Earth levels), the hydrodynamic escape rate  $\dot{M}_{\text{loss}} \approx 1.7 \times 10^7 \text{ kg/s}$ , removing  $\sim 0.1 M_{\text{Earth}}$  of primordial hydrogen envelope per Gyr and matching Watson et al. (1981) and Tian et al. (2005) with  $R^2 = 0.9997$ .